Operators and Datatypes in C

Operators

Here's a concise list of all the operators in C:

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1. Arithmetic Operators

- + (Addition)
- - (Subtraction)
- * (Multiplication)
- / (Division)
- % (Modulus)

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2. Relational Operators

- == (Equal to)
- != (Not equal to)
- > (Greater than)
- < (Less than)
- >= (Greater than or equal to)
- <= (Less than or equal to)

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3. Logical Operators

- && (Logical AND)
- || (Logical OR)
- ! (Logical NOT)

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4. Bitwise Operators

- & (Bitwise AND)
- | (Bitwise OR)
- ^ (Bitwise XOR)
- ~ (Bitwise NOT)
- << (Left shift)
- >> (Right shift)

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5. Assignment Operators

- = (Assignment)
- += (Add and assign)
- -= (Subtract and assign)
- *= (Multiply and assign)
- /= (Divide and assign)
- %= (Modulus and assign)
- &= (Bitwise AND and assign)
- |= (Bitwise OR and assign)
- ^= (Bitwise XOR and assign)
- <<= (Left shift and assign)
- >>= (Right shift and assign)

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6. Increment and Decrement Operators

- ++ (Increment)
- -- (Decrement)

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7. Conditional (Ternary) Operator

- ?: (Condition ? expr1 : expr2)

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8. Sizeof Operator

- sizeof (Returns size of a variable or data type)

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9. Pointer Operators

- & (Address-of)
- * (Dereference)

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10. Comma Operator

- , (Separates expressions in a single statement)

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11. Type Casting Operators

- (type) (Explicit type casting)

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12. Miscellaneous Operators

- -> (Access members of a struct through a pointer)
- . (Access members of a struct)
- [] (Array subscript)
- , (Comma for multiple expressions in a statement)

✅ **Summary:**

These operators are fundamental in C for performing operations on variables and constants.

Datatypes

Here’s a concise list of data types in C and their typical sizes:

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1. Basic Data Types

- **char**: 1 byte
- **short**: 2 bytes
- **int**: 4 bytes
- **long**: 4 bytes (varies based on system, can be 8 bytes on 64-bit systems)
- **long long**: 8 bytes
- **float**: 4 bytes
- **double**: 8 bytes
- **long double**: 8-16 bytes (varies by system)

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2. Derived Data Types

- **pointer**: 4 bytes (32-bit) or 8 bytes (64-bit system)
- **array**: Size depends on the element type and number of elements
- **struct**: Size depends on the member types and padding
- **union**: Size equals the size of the largest member
- **enum**: Typically 4 bytes (can be 2 bytes depending on the system)

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3. Type Qualifiers

- **const**: No size (qualifier modifier, used with other types)
- **volatile**: No size (qualifier modifier, used with other types)

✅ **Summary:**

The size of data types can depend on the system and compiler, but the typical sizes listed here are the most common.